Task 1 — Problem Definition & Scope

Agricultural Market Price Tracker (AMPT)

1. Problem Statement

Farmers, traders and buyers in local agricultural markets (mandis) currently rely on word-of-mouth, phone calls, or a physical visit to the market yard to find out the day's price for a commodity. Prices for the same crop can vary 15–40% between nearby mandis on the same day, but there is no single, low-friction place where a farmer can check current and historical prices, compare markets, and get warned when a price moves sharply before deciding where and when to sell.

The Agricultural Market Price Tracker (AMPT) is a web application that lets authorized users record daily commodity price data for a market, and lets any user search, view, and drill into that data through a dashboard with summary indicators and alerts for unusual price movements.

2. Target Users

User type	Need
Farmer / seller	Check today's price and recent trend before choosing where to sell
Trader / commission agent	Enter daily arrival & price data for their market; compare markets
Mandi / market official	Publish official price data quickly; see exceptions needing review
Consumer / buyer	Look up retail-relevant wholesale price trends
Government / agri department (secondary)	Monitor price volatility across markets

3. Existing Pain Points

1. Price discovery depends on informal networks — asymmetric information favors middlemen.
2. Manual, paper-based price recording at mandis; no searchable digital record.
3. No easy way to compare the same commodity across multiple markets.
4. No alerting when a price spikes or crashes — farmers find out too late.
5. Historical trend data, where it exists, is siloed in spreadsheets or PDFs.

4. Stakeholders

- **Primary users:** Farmers, traders/commission agents, mandi data-entry officials
- **Secondary users:** Consumers, agri-department analysts
- **Project stakeholders (for this academic/DevOps exercise):** Developer (you), course evaluator/instructor (acceptance of deliverables), "product owner" role (played by the developer to prioritize backlog)
- **Operations stakeholder:** Whoever runs the Jenkins/Docker/Ansible pipeline in later tasks (same person, wearing a DevOps hat)

5. Objectives

1. Give any user a searchable, up-to-date view of commodity prices by market and date.
2. Let authorized users enter/update price and arrival-quantity data for a market-day.
3. Surface at-a-glance summary indicators (e.g., today's average price, % change vs previous day, number of markets reporting).
4. Allow drill-down from a summary indicator into the underlying records.
5. Flag exceptions/alerts automatically (e.g., price change beyond a threshold, missing data for a market that usually reports).
6. Demonstrate a complete, automated DevOps pipeline (Git → CI → test → containerize → provision → deploy) around this application.

6. Constraints

- **Academic/solo-project constraints:** single developer, limited timeline, local or low-cost infrastructure (local Jenkins, Docker Desktop/local registry, Vagrant/local VM or containers for Ansible/Puppet targets — not a paid cloud requirement).
- **Technical constraints:** must run on a conventional Java or similar stack deployable to Tomcat/Nginx (per Task 3), fronted by a relational or lightweight database.
- **Data constraint:** no live government market-price API is assumed available by default; MVP uses manually entered / seeded data (an optional stretch goal can wire in a public data source later).
- **Time-boxing constraint:** MVP is deliberately frozen at 15 build tasks (Section 8) to keep scope demonstrable within the project timeline.

7. Measurable Success Criteria

#	Criterion	Target
1	Core user journeys implemented	5 of 5 (entry, dashboard, summary, drill-down, alerts)
2	Automated Selenium tests	3–5 critical journeys, passing in CI
3	CI build	Every commit to `develop`/`main` triggers a Jenkins build automatically
4	Deployment automation	One-command / one-pipeline deploy to a container, no manual server steps
5	Provisioning idempotency	Second Ansible/Puppet run reports zero changes
6	Rollback	Documented rollback to previous release succeeds in a test run
7	Search/dashboard responsiveness	Search result or dashboard load in local testing < 2s with seed data
8	Alert accuracy	100% of seeded "spike/crash" test records correctly flagged

8. Frozen MVP Scope — 15 Build Tasks

The application MVP is scoped to exactly the 5 functional areas required by Task 3, broken into 15 concrete, sequenced build tasks. This list is the backlog seed for Task 2 and will not grow during MVP development (later ideas go to a "Future Enhancements" backlog instead — see Task 15).

Foundation

1. Define data model: `Market`, `Commodity`, `PriceRecord` (date, market, commodity, min/max/modal price, arrival qty), `User`/`Role`
2. Set up project skeleton, build tool config, and local DB schema/migration

Data/Event Entry 3. Price-entry form (create) with server-side validation 4. Edit/correct an existing price record (update) 5. Basic role check: only authorized roles can enter/edit data

Searchable Dashboard 6. Search API + UI: filter by market, commodity, date range 7. Paginated results list with sort (by date/price)

Summary Indicators 8. Compute today's average/modal price per commodity 9. Compute % change vs previous reporting day 10. Dashboard widget row showing the above indicators

Status Drill-Down 11. Click-through from a summary indicator to its underlying record list 12. Single-record detail view (full history for one market+commodity)

Alert / Exception View 13. Rule: flag a record when price change exceeds a configurable % threshold 14. Rule: flag a market as "missing today's data" if no record by a cutoff 15. Alerts/exceptions list view, separate from the main dashboard

Each of these 15 items becomes a backlog item with acceptance criteria in Task 2, and the 5 functional-area groupings above map directly to the feature branches built in Tasks 5–6.

9. Out of Scope for MVP (Future Enhancements)

- Live integration with a government/public market-price data feed
- SMS/WhatsApp push alerts (in-app alert view only for MVP)
- Multi-language UI
- Mobile app (responsive web only)
- Role-based analytics dashboards for government stakeholders

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Deliverable status: Task 1 — approved for use as the baseline for Task 2 (backlog & sprint plan).